



Principles of Radiographic Interpretation

Dr.Manila

OMFS 661

Mission: To advance the science and art of healthcare through education, service, research, personal wellness, and social accountability.

Learning Objectives & Competency Statements

Learning Objectives At the end of this course students will be able to:	CDM/CODA Competency #s	Teaching Activity (Lecture, Lab, Small group, etc.)	Method of Assessment
Understand the principles of radiographic imaging in the context of oral and maxillofacial pathology, including the etiology, pathogenesis, clinical features, and histopathology characteristics of relevant conditions.	CDM: 1.1, 1.3, 2.2, 3.1, 3.2, 3.3, 4.1, 5.1, 5.2, 6.1, 6.2 CODA: 2-10, 2-14, 2-15, 2-17, 2-23, 2-24	Lecture, Small group learning, Case-based Learning	Small Group Feedback, Quizzes, Midterm Examination, Final Examination
Apply the knowledge of normal oral and maxillofacial anatomy to recognize and diagnose oral pathologies on radiographic images.	CDM: 1.1, 1.3, 2.2, 3.1, 3.2, 3.3, 4.1, 5.1, 5.2, 6.1, 6.2 CODA: 2-10, 2-14, 2-15, 2-17, 2-23, 2-24	Lecture, Small group learning, Case-based Learning	Small Group Feedback, Quizzes, Midterm Examination, Final Examination
Apply the knowledge of basic principles of different imaging modalities used in oral and maxillofacial radiology and how these principles can be applied to the diagnosis of various pathologies.	CDM: 1.1, 1.3, 2.2, 3.1, 3.2, 3.3, 4.1, 5.1, 5.2, 6.1, 6.2 CODA: 2-10, 2-14, 2-15, 2-17, 2-23, 2-24	Lecture, Small group learning, Case-based Learning	Small Group Feedback, Quizzes, Midterm Examination, Final Examination
Demonstrates behaviors and values that are consistent with the trust given to the profession by patients, other healthcare providers, and society.	CDM: 2.1 CODA: 2-21	In-class participation and activities	EPA



Week/ Session	Dates	Topic	Reading Assignment	Assignments and Assessments	Faculty
1	7/13/2026	Principles of Radiographic Interpretation	Rad: Mallya 8 th -Ch. 18	Activity EPA	Manila
2	7/20/2026	Dental Caries and Periodontal Diseases	Rad: Mallya 8 th -Ch. 19, 20	Activity EPA	Manila
3	7/27/2026	Inflammatory Conditions of the Jaws	Rad: Mallya 8 th -Ch. 22 Path: Neville 5 th – Ch. 3	Activity EPA	Manila Wiley
4	8/3/2026	Cysts	Rad: Mallya 8 th -Ch. 23 Path: Neville 5 th – Ch. 1, 15	Activity Quiz EPA	Manila Wiley
5	8/10/2026	Benign Tumors and Neoplasms	Rad: Mallya 8 th -Ch. 24 Path: Neville 5 th – Ch. 14, 15	Activity EPA	Manila Wiley
6	8/17/2026	Diseases Effecting Structures of Bone	Mallya 8 th -Ch. 25	Activity EPA	Manila Wiley
7	8/24/2026	Paranasal Sinus Abnormalities	Rad: Mallya 8 th -Ch. 28	Activity Quiz EPA	Manila
8	8/31/2026	Midterm		Midterm	Manila Wiley
9	9/7/2026	Holiday			
10	9/14/2026	Malignant Neoplasms	Rad: Mallya 8 th -Ch. 26 Path: Path: Neville 5 th – Ch. 14	Activity EPA	Manila Wiley
11	9/21/2026	Trauma - Guest lecture	Mallya 8 th -Ch. 27	Activity Quiz EPA	Dr. Christensen OMFS
12	9/28/2026	Soft Tissue Calcifications and Ossifications	Rad: Mallya 8 th -Ch. 31	Activity EPA	Manila
13	10/5/2026	TMJ Abnormalities -Guest lecture	Rad: Mallya 8 th -Ch. 30	Activity EPA	Dr. Salah OFP
14	10/12/2026	Differential Diagnosis/ Activity	Path: Course Presentation	Activity Quiz EPA	Wiley
15	10/19/2026	Finals Week		Finals	Manila/Wiley

What is Expected of You

- Stay focused and attentive during class.
- Treat others with respect at all times.
- Approach class with the same responsibility and professionalism as you would a workplace.
- Communicate thoughtfully and respectfully.
- Maintain a professional attitude in all class interactions.
- Be punctual and bring a growth mindset and curiosity to each session.
- Establish and maintain a consistent study schedule.
- Recording of any kind (including photos of activities) is not permitted.
- Sitting near the front is encouraged to enhance your learning experience.
- Share any concerns or issues with me in advance whenever possible.
- Review and stay familiar with the syllabus available on Canvas.



Attendance and Professional Expectations

- Attend every class unless you have an approved absence.
- Arrive on time and remain in class for the entire scheduled session.
- Do not schedule personal appointments during class time unless they are unavoidable. If an unavoidable appointment requires you to miss class or leave early, communicate with the instructor as soon as possible and follow the required absence procedures.
- Extended absences from the classroom during class are not acceptable except in extenuating circumstances.
- If you need to be absent for any reason, you are responsible for submitting the required absence request in a timely manner.
- Faculty should not have to send multiple reminders requesting the appropriate documentation.
- It is your responsibility to stay informed about course expectations, policies, and any material missed due to an absence.



Interpretation

- An explanation of what is viewed on dental radiograph
- The ability to read what is revealed in dental radiograph



Diagnosis

Identification of disease by examination or analysis

Objectives of radiographic interpretation

3D

Objectives of radiographic interpretation

- Detection: Presence or absence of disease
- Description: providing information on the disease according to systematic approach
- Differentiation: diseases of similar radiographic appearance



Importance of Radiologic Interpretation

- Dentists must master:
 - Perception: recognizing abnormal patterns
 - Cognition: interpreting those patterns
- These are inseparable components of visual diagnosis.

Clinical examinations

- Radiographs are prescribed when the dentist thinks that they are likely to offer useful diagnostic information.
- Often some clinical sign or symptom or finding from the patient's history indicates the need for a radiological examination.
- This clinical information should be used first to select the type of radiographs.

Acquiring appropriate diagnostic images

- An **insufficient number, inadequate quality** of radiographs limits the information available from diagnostic imaging.
- Because the General practitioner often is responsible both for prescribing and interpreting radiographs.
- Inadequate radiographs should be recognized, and **supplemental images** must be obtained before proceeding with the analysis.



Essential Requirements For Interpretation

- Optimum viewing conditions
- Knowledge of what the radiograph should look like
- Knowledge of appearance of normal anatomical landmarks as well as variations of normal
- Knowledge of radiographic appearance of pathological conditions affecting the maxillofacial region.



Radiology Reading Room

Quality of diagnostic X rays

- Conventional dental radiography produces images in only 2 dimensions., usually in the mesiodistal direction.
- In some cases, view at right angles to the plane of the original film is beneficial.
- Diagnostic imaging should be completed before a biopsy procedure or treatment is provided.
- Diagnostic imaging can aid in the selection of the most appropriate site for biopsy procedure to obtain the most representative tissue sample.

Systematic description and analysis of findings revealed in the radiograph

Image analysis

- Systematic radiographic examination
 - The first step is to use a systematic approach to identify all the normal anatomy present in an image or set of images.
 - A profound knowledge of the variation of normal appearance is required to be able to recognise an abnormal appearance. The best learning method is to identify normal anatomy in every image analysed.
 - Practitioners should avoid limiting their attention to one particular region of the radiograph.
 - All aspects of each image should be examined systematically.
 - More than one abnormality may be present.



Intraoral Imaging

- Periapical and bitewing X rays: crown, root, pulp, PDL space, lamina dura
- Right Maxilla. Left Maxilla. Left mandible, right mandible.
 - Bone and quality of trabecular bone.
 - Teeth.
 - Restorations.

Extraoral Radiography

- Most commonly used in dentistry are panoramic images, cephalometric views and examinations of the temporomandibular joints.
- Panoramic: maxilla border, sinus floor, zygomatic process
- CBCT: review axial, coronal, sagittal planes
- It is important to develop a definitive sequence that considers all the hard and soft tissues in the field.



Importance of Normal Anatomy Knowledge

- Recognizing normal variations is key to identifying abnormalities
- Especially critical in panoramic images
- Enhances meaningful image review



Diagnostic Reasoning Process

- Unique to oral/maxillofacial radiology:
- Differentiate normal vs abnormal in 2D images
- Form 3D mental representation of abnormalities



Mental Representation of Abnormalities

- Includes:
 - Location
 - Size
 - Internal structure
 - Effects on surrounding structures



Ineffective Novice Strategy

- Memorizing features of abnormalities is ineffective
 - Leads to poor diagnostic accuracy
 - Lacks understanding of disease mechanisms



Effective Strategy - Disease Mechanisms

- Understanding pathophysiology improves accuracy
 - Biochemical, cellular, tissue, and organ level changes
 - Helps form coherent mental models

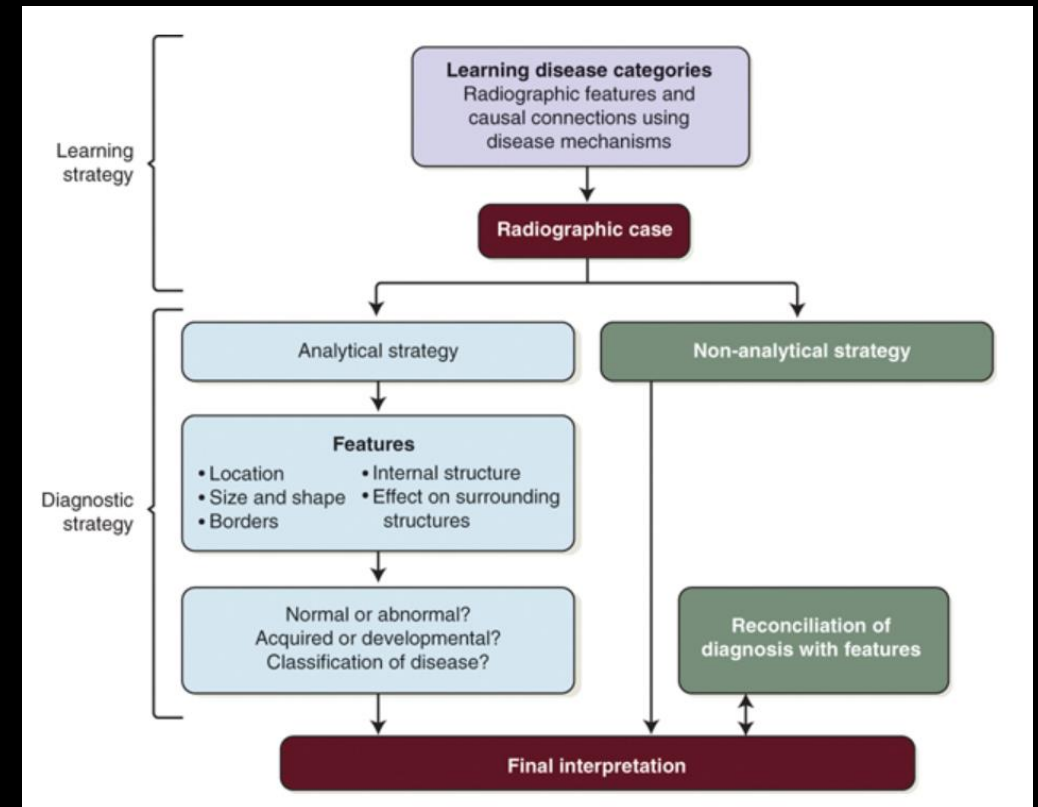


Worth's Principle

" Radiographic appearances are governed by anatomic and physiologic changes in the presence of disease processes. Radiologic diagnosis is founded on knowledge of these alterations, the prerequisite being awareness of disease mechanisms."

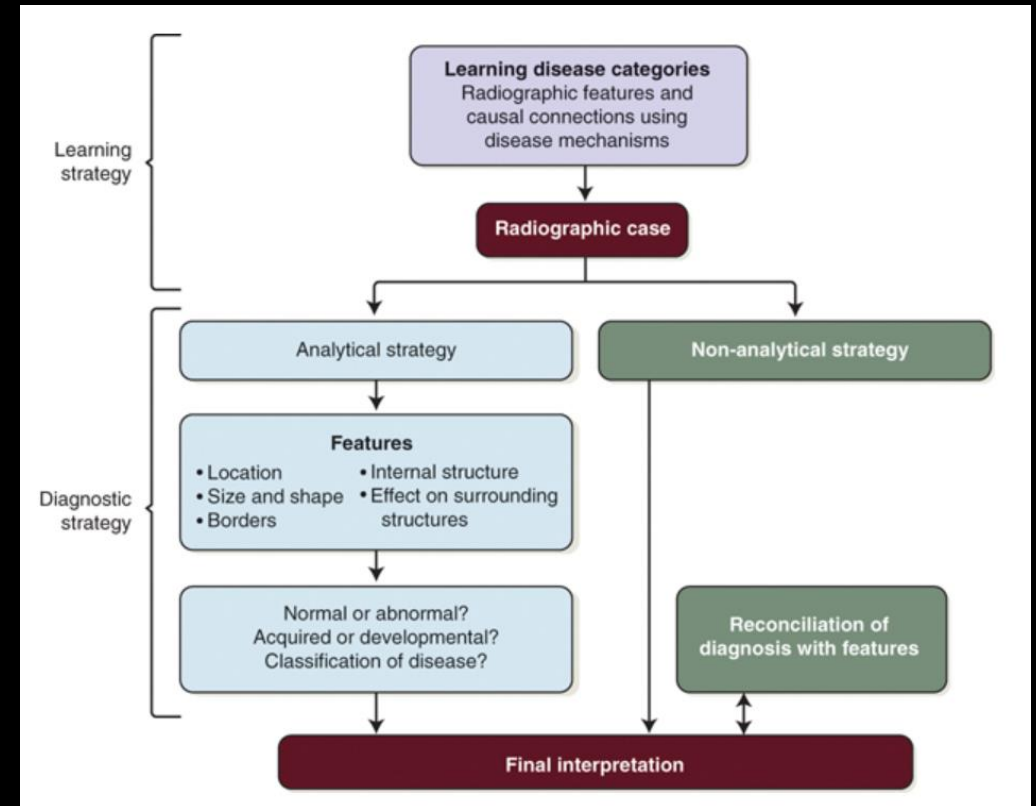
Analytic Diagnostic Strategy

- Step-by-step analysis of features:
 - Reduces bias
 - Prevents premature conclusions
 - Encourages thorough evaluation



Nonanalytic Strategy

- Pattern recognition approach:
 - Holistic view of abnormality
 - Forms hypothesis, then searches for supporting features

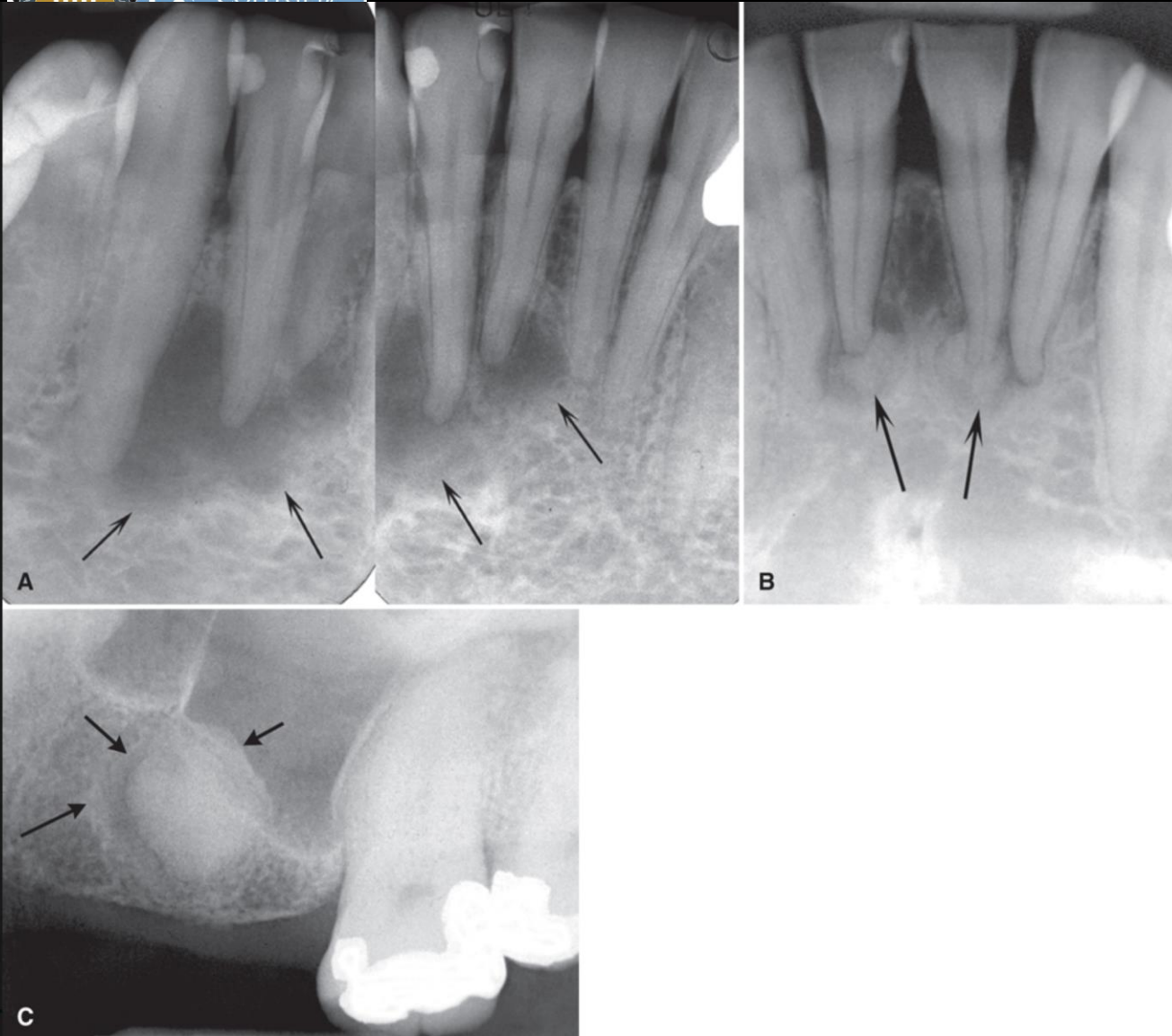


Analysis of abnormal findings

Two basic approaches

1. Picture matching or aunt minnie method.

- This involves trying to match the radiographic image with a mental picture or with an image in a favourite textbook.
- Limitations: The observers experience and memory and the appearance of different abnormalities. Also, figures in a textbook usually represent the most ideal example of the abnormality.
- For example, the term periapical cemental display share often evokes mental image of radiolucent or a mixed radiolucent radiopaque lesion at the apex of mandibular incisor.



Series of periapical images showing different maturation phases of cemento-osseous dysplasia.

- (A) Early radiolucent phase after periapical bone has been resorbed and replaced with a connective tissue matrix.
- (B) Late maturation phase (arrows) showing central amorphous bone (radiopaque) surrounded by a radiolucent internal border.
- (C) Mature phase (arrows) of cemento-osseous dysplasia in an unusual location and after the associated tooth has been extracted.

Analysis of abnormal findings

2. Systematic approach

- Step 1: Localise the Lesion.
- Step 2: Assess the periphery and shape.
- Step 3. :Analyse the internal structure.
- Step 4. :Analyse the effect on surrounding structure.
- Step 5. :Formulate a radiographic interpretation.

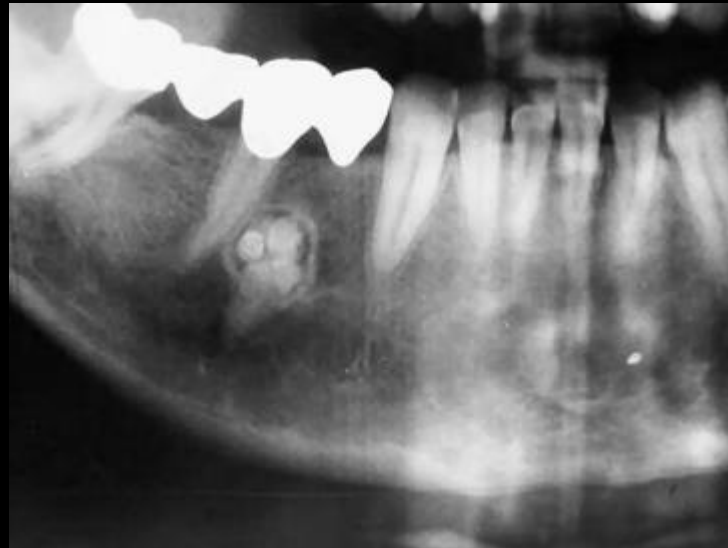
Localise the abnormality

- Localised or Generalised?
 - Many abnormalities are localised to a specific region.
 - If an abnormal appearance affects all the osseous structures of the maxillofacial region, generalised conditions such as metabolic or endocrine abnormalities of the bone are considered.
 - If there abnormality is localised, it may be unilateral or bilateral.
 - Often variations of normal anatomy are more commonly bilateral, whereas abnormal conditions are more commonly unilateral.
 - Fibers display share commonly is unilateral.
 - However, a few abnormalities such as Paget's disease, cherubism are always seen bilaterally in the Mandible.



Cherubism, is bilateral, manifesting in both the left and the right mandibular rami. Because the origin of the lesion is in the midramus region, the mandibular molars have been bodily displaced mesially toward the anatomical midline on both sides.

Localized

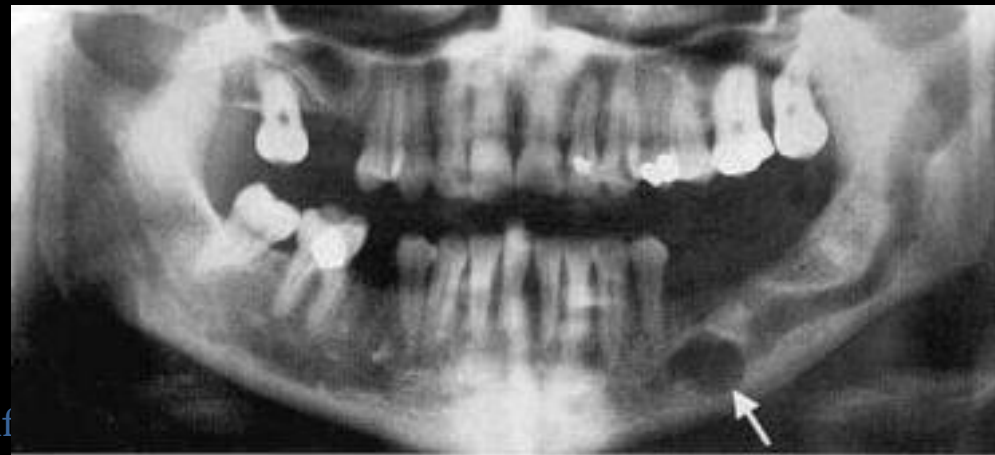


Generalized



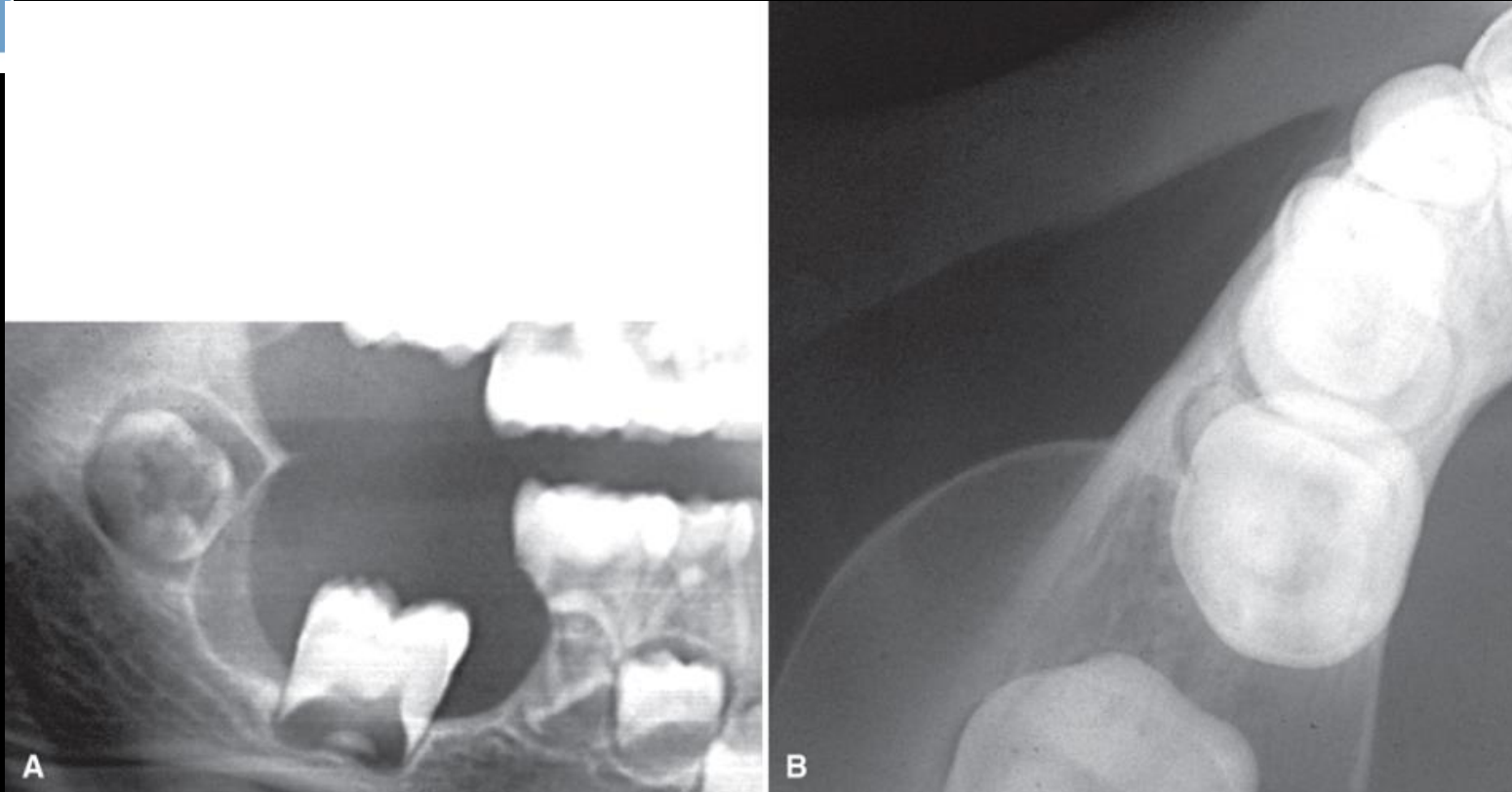
Unilateral Bilateral





Localise the abnormality

- Position in the jaws
 - Is the abnormality in soft tissue or it is contained within the jaws/hard Tissue?
 - When the lesion is in bone, the point of origin or epicenter?
 - Determining the exact location may be difficult in some circumstances if the nature of their normality is not well defined.
- The following are a few examples.
 - If the epicenter is coronal to a tooth, the lesion probably is composed of odontogenic epithelium.
 - If it is above the inferior alveolar nerve canal the likelihood is greater that it is composed of odontogenic tissue.
 - If the epicenter is below IAC it is unlikely to be Odontogenic in origin.
 - If it originates within IAC, the tissue of origin probably is neural or vascular in nature.



(A) Cropped panoramic image of a lesion where the epicenter is located coronal to the unerupted mandibular first molar. (B) Occlusal projection providing a right-angle view of the same lesion.



Panoramic image of an ameloblastoma in the body of the left mandible. The inferior alveolar nerve canal has been displaced inferiorly to the inferior cortex (arrows), indicating that the lesion started superior to the canal.



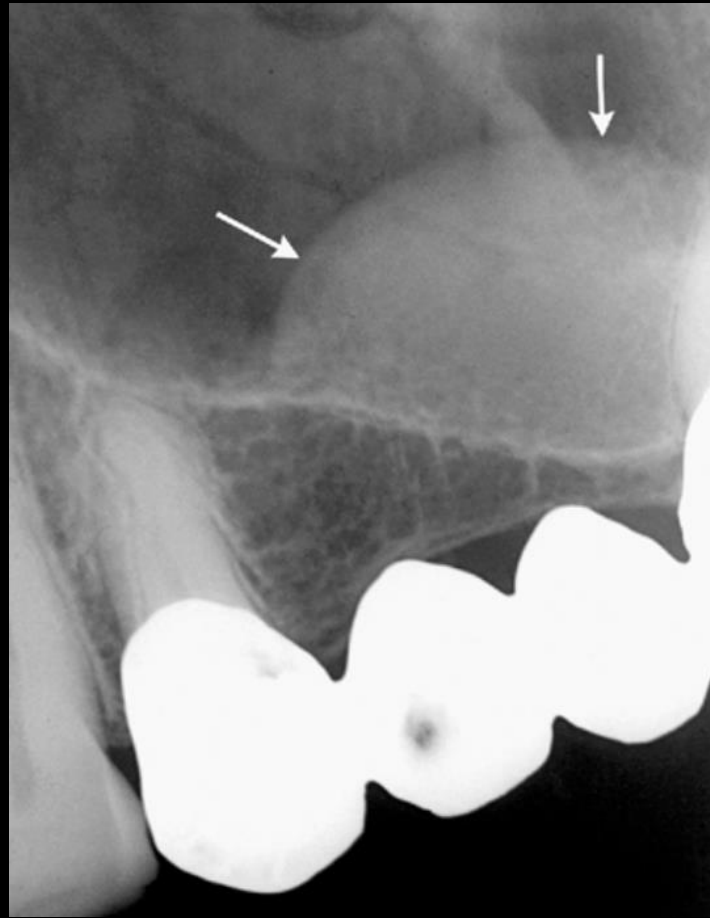
Cropped panoramic image displaying a lesion (a submandibular salivary gland or Stafne defect) with an epicenter located inferior to the inferior alveolar canal. This characteristic is unlikely to be associated with an odontogenic origin.



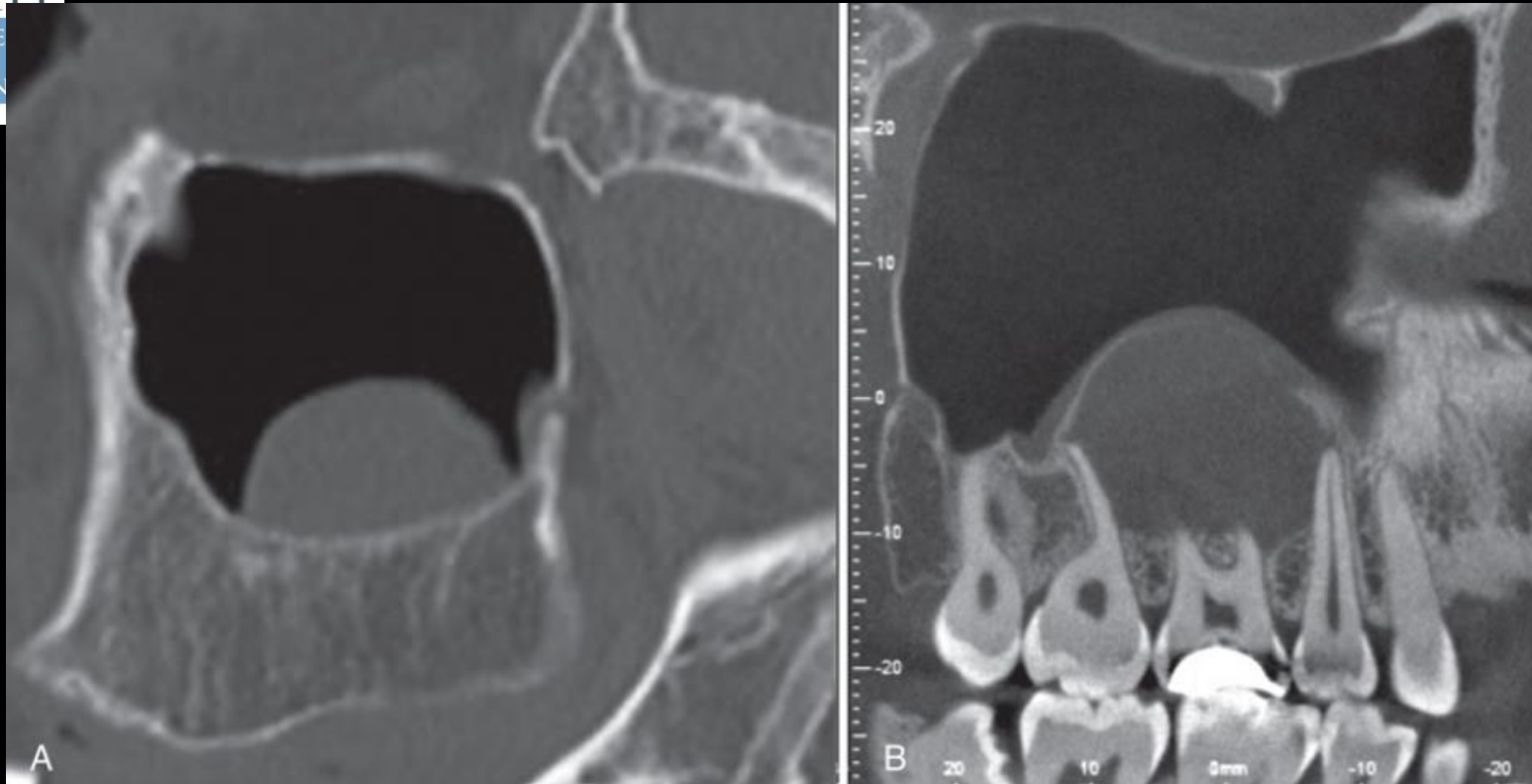
Lateral oblique image of the mandible revealing a lesion within the inferior alveolar canal. The smooth fusiform expansion of the canal suggests a neural lesion.

Localise the abnormality

- The probability of cartilaginous lesions and osteochondromas occurring is greater in condylar region.
- If the epicentre is within the maxillary antrum, the lesion is not of odontogenic tissue, as opposed to a lesion that has grown into the end from the alveolar process of the Maxilla.
- Particularly, lesions tend to be found in specific locations.
 - The epicentres of central giant cell granulomas commonly are located anterior to the first molars in the mandible and anterior to the cuspid in the maxilla in young patients.
 - Osteomyelitis occurs in the mandible and rarely in the maxilla.
 - Periapical cemental dysplasia occur in the periapical region of teeth.



The lack of a peripheral cortex (arrows) on this retention pseudocyst indicates that it originated from within the sinus and not in the alveolar process. Therefore, it is unlikely to be of odontogenic origin.



(A) Sagittal cone beam computed tomographic image of the maxillary sinus. The lack of a peripheral cortex on this retention pseudocyst and the preservation of the floor of the sinus indicates that it has originated in the sinus and not the alveolar process. (B) In comparison, an odontogenic cyst originating in the alveolar process, displacing the floor of the maxillary sinus superiorly, exhibits a cortex surrounding the lesion.

Localise the abnormality

- Single or multi focal
 - The list of possible multifocal abnormalities is short.
 - Some examples are Periapical Cemental dysplasia, OKC, metastatic lesions, multiple myeloma, and leukemic infiltrates.
- Size
 - There are very few size restrictions for a particular region, but the size may aid in the differential diagnosis.
 - For instance, a dentigerous cyst is often larger than a hyperplastic follicle.

Assess the periphery and shape

- Study the periphery of the lesion, well defined or ill defined.
- If an imaginary pencil can be used to draw confidently the limits of the lesion, the margin is well defined.
- If some small regions are ill defined these may be due to the shape or direction of the X-ray beam.
- A well-defined lesion is one in which most of the periphery is well defined, traceable.

Assess the periphery and shape

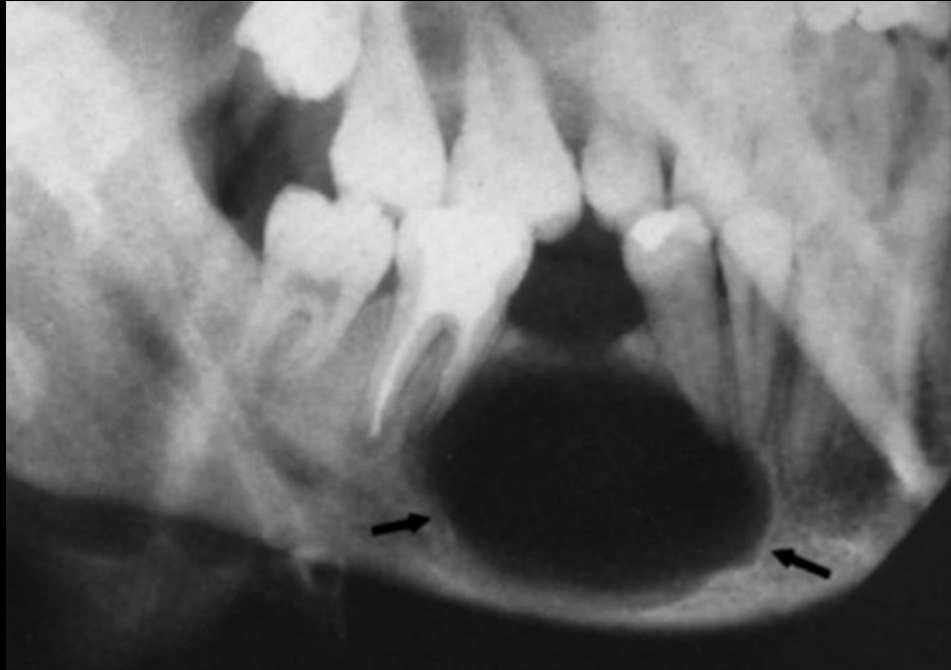
- Well-defined borders suggest slow-growing or benign process
- A punched-out border is one that has a sharp boundary in which no bone reaction is apparent immediately adjacent to the abnormality.
 - This is analogous to punching a hole in a radiograph with the paper punch; Multiple myeloma.
- A corticated margin is a thin, fairly uniform radiopaque line of reactive bone at the periphery of lesion; cysts.
- Sclerotic margin is a wider radiopaque border of reactive bone that usually is not informing width , periapical cemental dysplasia

Assess the periphery and shape

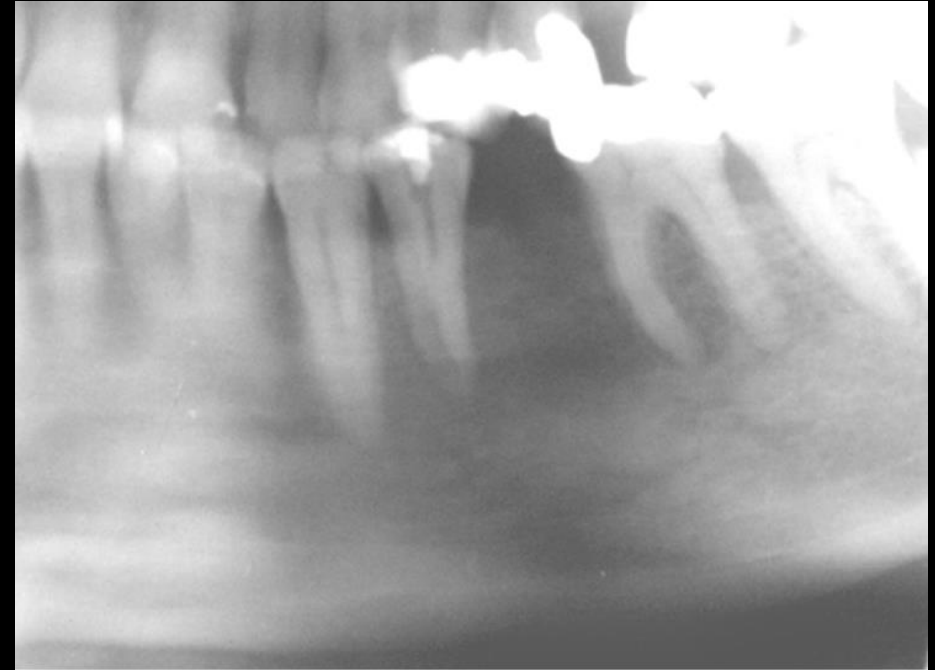
- A radiopaque lesion may have a soft tissue capsule which is indicated by the presence of radiolucent line at the periphery; Odontomas and cementoblastomas.

Assess the periphery and shape

- Ill-defined borders
 - A gradual transition between normal appearing bone trabeculae and the abnormal appearing trabeculae of the lesion.
 - Some conditions with this type of margin are sclerosing osteitis and fibrous dysplasia.
 - An invasive border usually is associated with the rapid growth and can be seen with the malignant lesions.
 - Usually an area of radiolucency with fewer or no trabeculae, representing bone destruction, can be seen just behind the margin.
 - These borders have also been described as permeative because the lesion grows around existing trabecular, producing radiolucent finger like or bay-type extensions at the periphery; may result in enlargement of the marrow space at the periphery.



Lateral oblique image of the mandible showing the well-defined border (arrows) of a residual cyst.

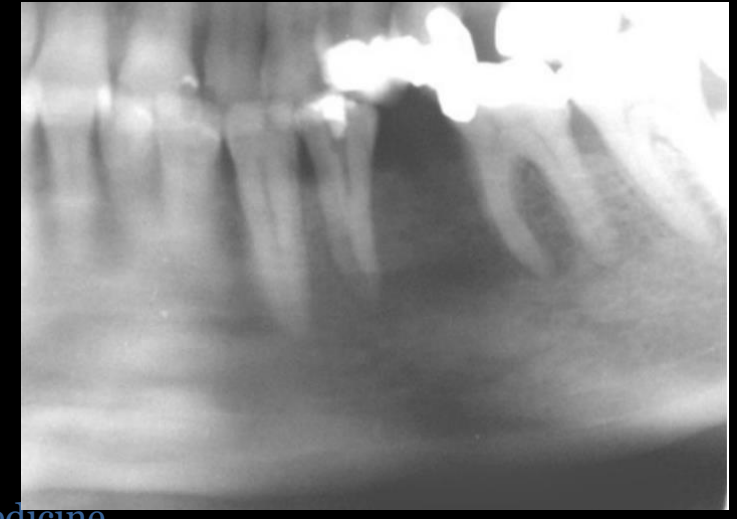
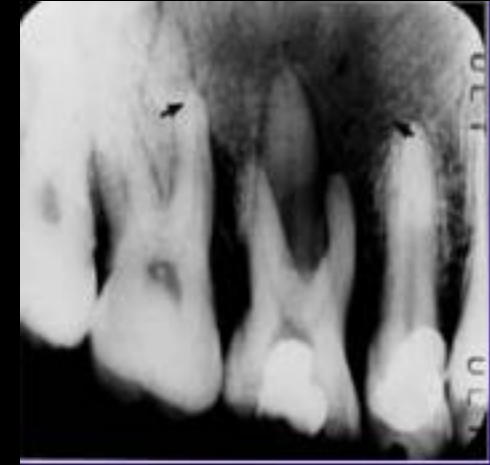


Cropped panoramic image showing the poorly defined border of a malignant neoplasm that has destroyed bone between the first molar and the first bicuspid.

- Border
– Well-defined



- Border
– Ill-defined



Assess the periphery and shape

- Shape
 - Lesion may have a particular shape, or it may be irregular.
 - The following are some examples.
 - A circular or fluid filled shape inflated balloon; as hydraulic, Cyst
 - A scalloped shape is a series of contiguous arcs or semicircles that may reflect the mechanism of growth. Cysts/OKC, Cyst like lesions/SBC, and some tumours.
 - Occasionally, a lesion with the scalloped periphery is referred to as multilocular internal structure.

Shape

- Round
- Oval
- Scalloped
- Irregular
- Inverted pear
- Heart shaped



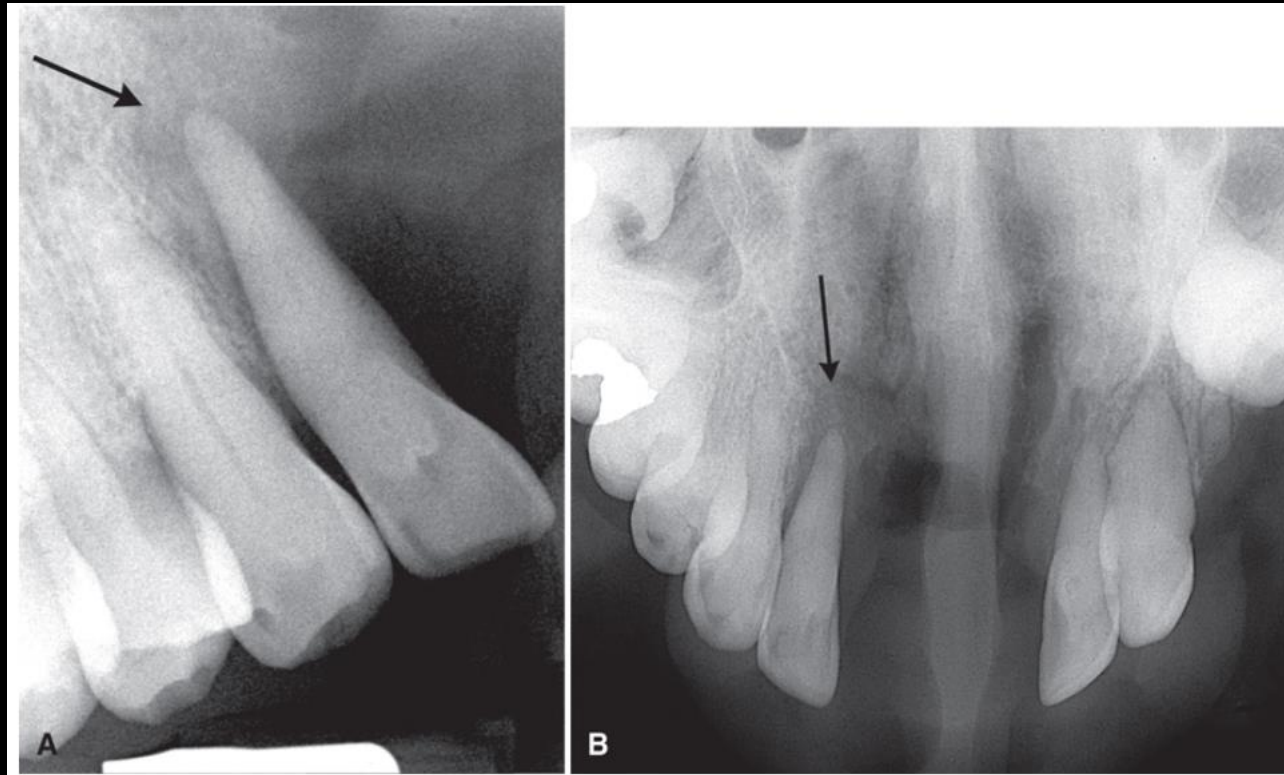
Thin, radiolucent periphery indicates a connective tissue capsule positioned between the internal radiopaque structure of this odontoma and the radiopaque outer cortical boundary (arrows).



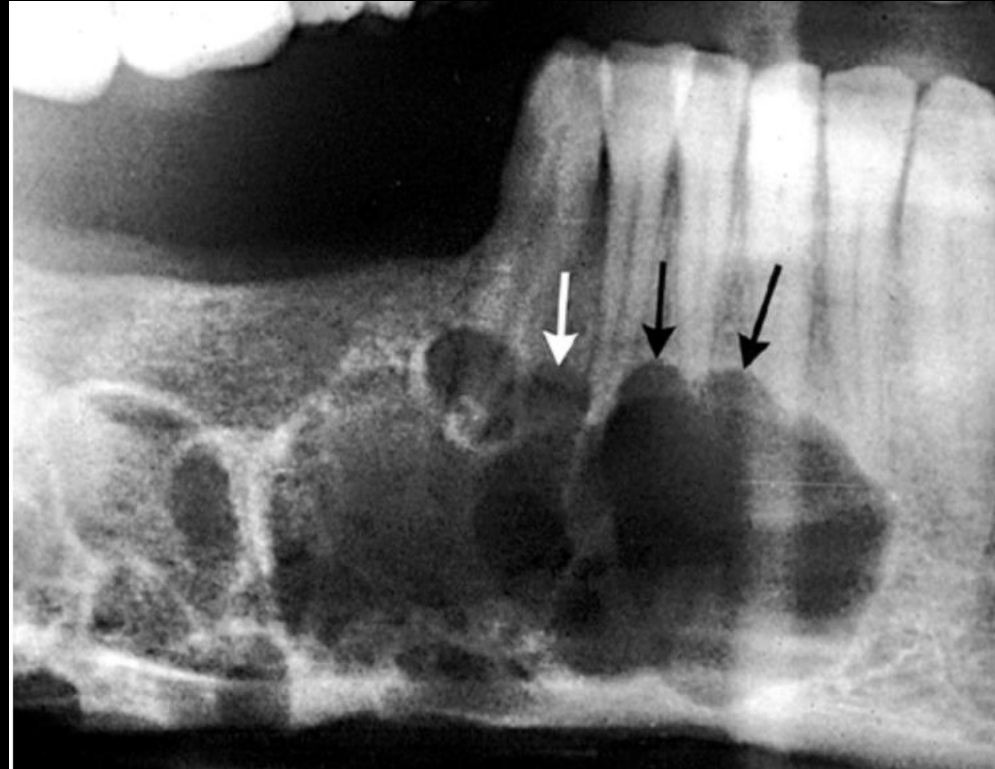
Periapical image revealing a radiopaque mass associated with the root of the first premolar. The prominent radiolucent periphery (arrows) is characteristic of a connective tissue capsule of this cementoblastoma.



Periapical image shows a gradual transition from the dense trabeculae of sclerosing osteitis (short arrow) to the normal trabecular pattern near the crest of the alveolar process (long arrow). This is an example of a poorly defined blending border.



Periapical (A) and occlusal (B) images reveal a squamous cell carcinoma in the anterior maxilla. The invasive margin extends beyond the lateral incisor (arrow), and the radiolucent region with no apparent trabeculae represents bone destruction behind this margin.



Cropped panoramic image of an odontogenic keratocyst displaying a scalloped border, especially around the apex of the associated teeth (arrows).

Analyse the internal structure

- Basic categories
 - Totally radiolucent
 - Totally radiopaque
 - Mixed
- A radiolucent interior is common in cysts.
- Totally radiopaque interior is observed in osteomas.
- Mixed density internal structure be seen as the presence of calcified structures against radiolucent backdrop.
 - Fibrous displays: The trabecular pattern usually are greater in number. Shorter and not aligned in response to applied stress to the bone, but are randomly oriented resulting in patterns described as have an orange-peel or ground-glass appearance.

Analyse the internal structure

- SEPTA represented residual trapped bone that has been organised into long strands of walls within the lesion.
 - If these septa divide the internal structure into at least 2 compartments, the term multilocular is used.
 - GCG: Wispy or granular , Odontogenic myxomas: straight thin septa.
- Dystrophic calcification occurs in damaged soft tissue. Calcified lymph nodes that appear as dense, cauliflower like masses in the soft tissue.
- Cementum usually has a homogeneous, dense, amorphous structure and sometimes is organised into round or oval shape.
- Tooth structure usually can be identified by the organization into enamel, dentin and pulp chambers. Also, the internal density is equivalent to tooth structure and greater than surrounding bone.

Analyse the internal structure

- List of most radiolucent to most radiopaque material seen in plain radiographs.
- Air, fat and gas.
- Fluid.
- Soft tissue.
- Bone marrow.
- Trabecular bone.
- Cortical bone and dentin.
- Enamel.
- Metal.

Effects on surrounding structures

- Space occupying lesions
 - Lesion that slowly creates its own space by displacing teeth and other surrounding structures.
 - Teeth, lamina, dura and periodontal membrane space.
 - More commonly with slow-growing space occupying lesions.
 - Lesions with an epicentre above the crown of tooth displace the tooth apically.
 - Lesion that start in the ramus, such as cherubism may push teeth in an anterior direction.
 - Lymphoma, leukemia, Langerhans cell histiocytosis grow in the papilla of developing teeth and may push the developing tooth in a coronal direction.

Effects on surrounding structures

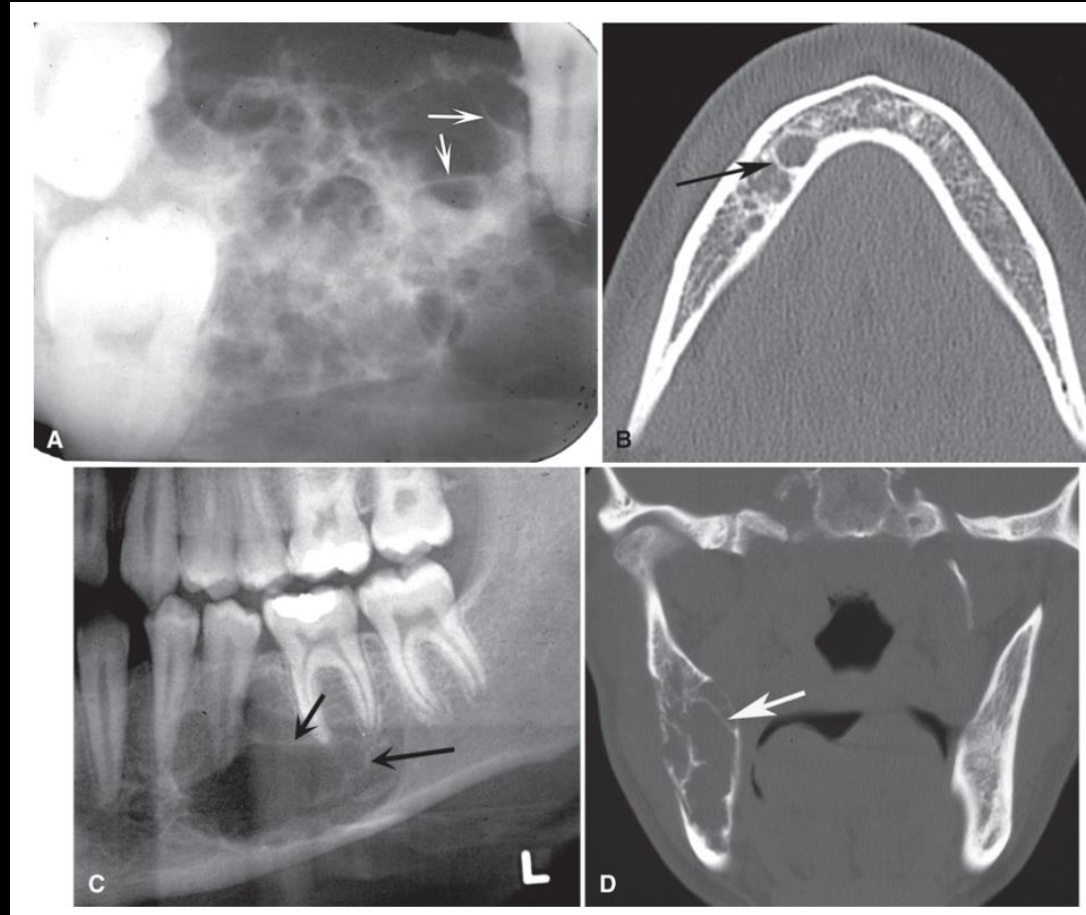
- PDL widening and lamina dura
 - Orthodontic movement of teeth causes widening of PDL space, but the lamina dura remains intact.
 - Malignant lesions can quickly grow down the ligament space resulting in an irregular widening and destruction of the lamina dura.
 - Resorption of teeth usually occurs with the more chronic or slowly growing process and may result from chronic inflammation.
 - Malignant tumours occasionally resorb teeth.

Effects on surroundings structure

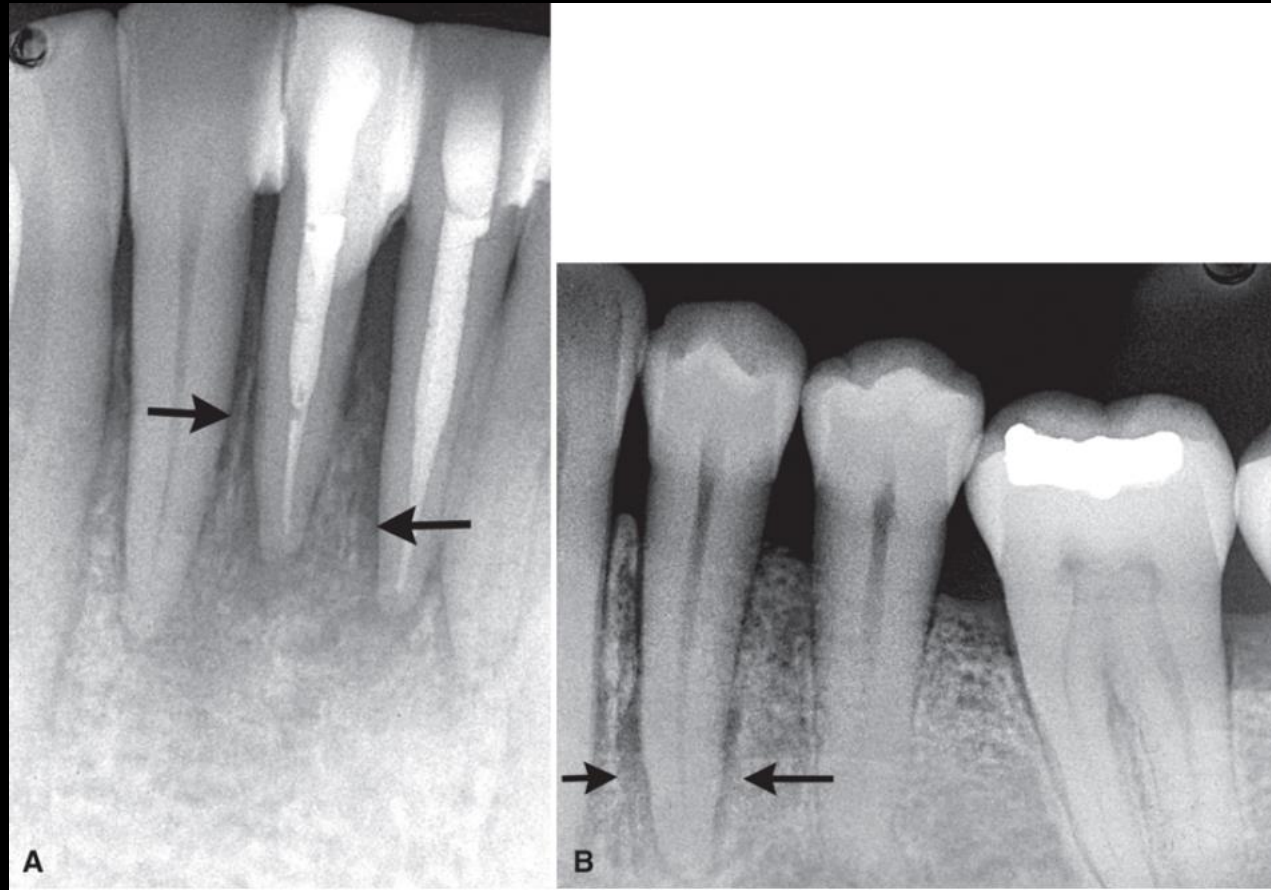
- Surrounding bone density and trabecular pattern.
 - The presence of reactive bone at the periphery of a lesion, whether corticated or sclerotic, usually signifies slow benign growth and possibly the ability to stimulate osteoblastic activity in the surrounding region.
- Inferior alveolar nerve canal and mental foramen.
 - Superior displacement is strongly associated with fibrous dysplasia
 - Widening of the IAC with the maintenance of a cortical boundary may indicate the presence of benign lesion of vascular or neural origin.
 - Irregular widening with the cortical destruction may indicate the presence of a malignant neoplasm growing down the length of the canal.

Effects on surroundings structures

- Outer cortical bone and periosteal reactions.
 - Exudate from an inflammatory lesion can lift the periosteum of the surface of the particle bone and then stimulate the periosteum to lay down new bone.
 - When this process occurs more than once and onion skin type of pattern inflammatory lesions and more rarely in some malignant lesions and in Langerhans cell histiocytosis.
 - Some periosteal reactions are very specific, such as speculated newborn formed at right angles to the outer cortical plate, Metastatic lesions of the prostate gland or in a radiating pattern in osteogenic sarcoma.



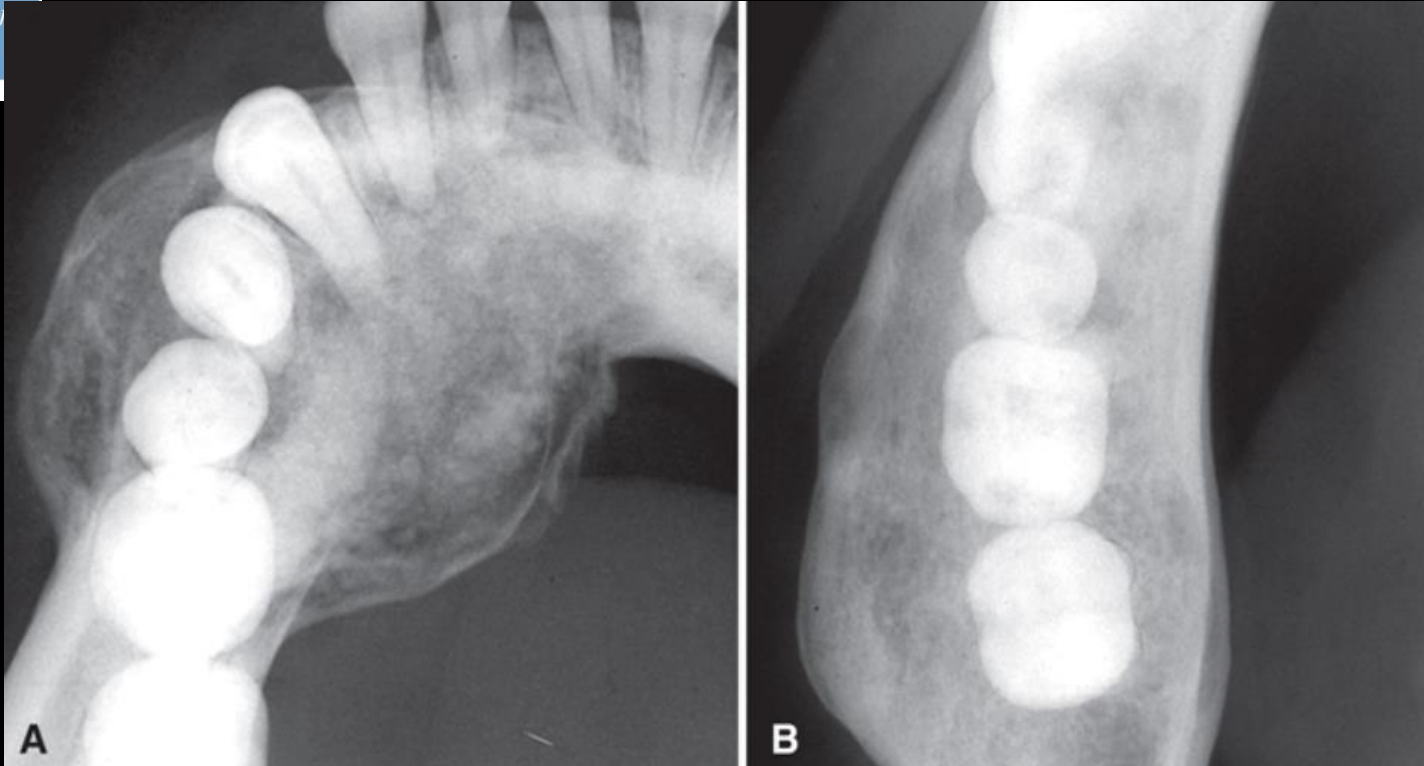
(A) Periapical image of an ameloblastoma. The multilocular pattern created by septa (arrows) divides the internal structure into smaller, soap bubble-like compartments. (B) Axial multidetector computed tomography image of an ameloblastoma has typically curved septa (arrow). (C) A cropped panoramic image of a central giant cell lesion with faint septations (arrows). (D) Coronal computed tomography image of an odontogenic myxoma with typically straight septa (arrow).



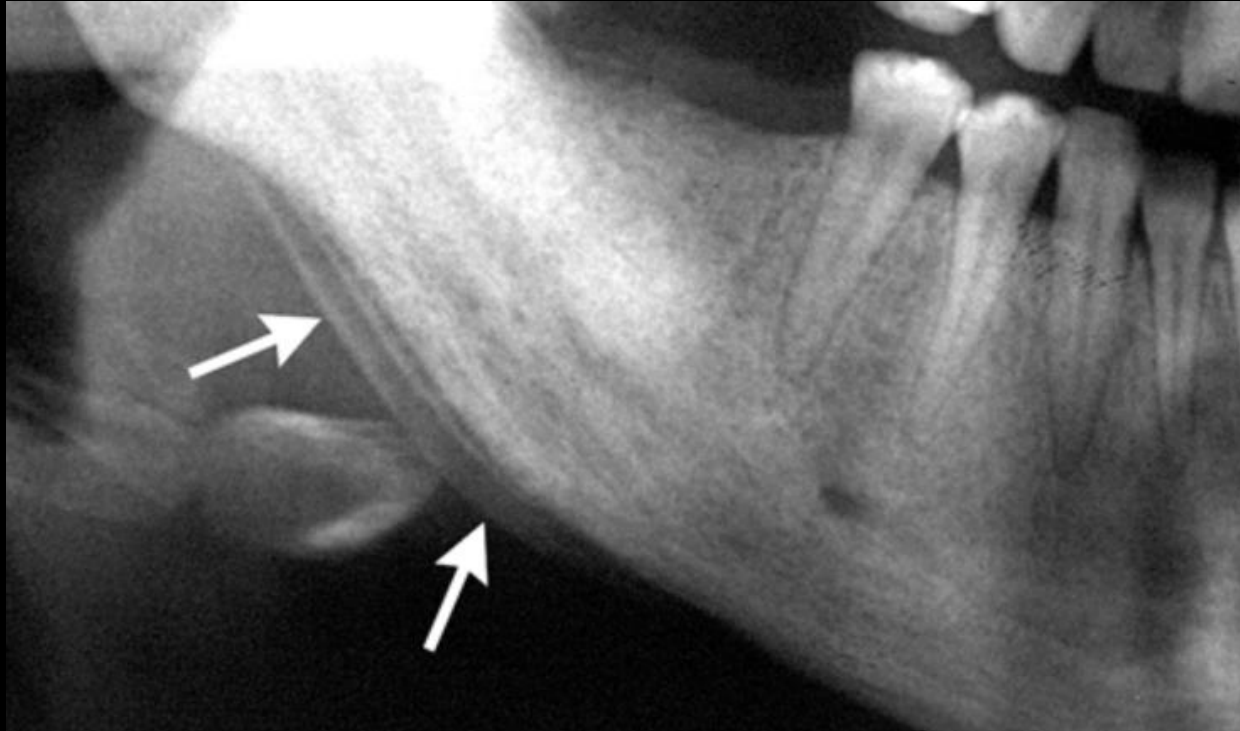
(A and B) Periapical images revealing a lymphoma that has invaded the mandible. There is irregular widening of the periodontal ligament spaces (arrows).



Axial computed tomography image of an ameloblastoma involving the left mandibular ramus shows significant expansion of the ramus with displacement of the bone border (black arrow). In some areas (white arrows), the speed of growth of the tumor has exceeded the ability of the bone to keep it confined.



(A) Occlusal image of a cemento-ossifying fibroma. The concentric expansion of the mandible is characteristic of a benign tumor. (B) Occlusal image of fibrous dysplasia with enlargement of bone along the length of the mandibular body without an obvious epicenter.



Panoramic image of osteomyelitis revealing at least two layers of new bone (arrows) produced by the periosteum at the inferior border of the mandible.



Specimen image of a resected mandible with an osteosarcoma. Note the fine linear spicules of bone at the superior margin of the alveolar process (arrows).

Systematic approach

Step 1: Localise the Lesion

- Anatomic position (epicenter)
- Localized or generalized
- Unilateral or bilateral
- Single or multifocal

Step 2: Assess the periphery and shape.

- Well defined
- Punched out
- Corticated
- Sclerotic
- Soft tissue capsule
- Poorly defined
- Blending
- Invasive
- Circular
- Scalloped
- Irregular

Step 3 :Analyse the internal structure

- Totally radiolucent
- Totally radiopaque
- Mixed radiolucent and radiopaque (describe pattern)

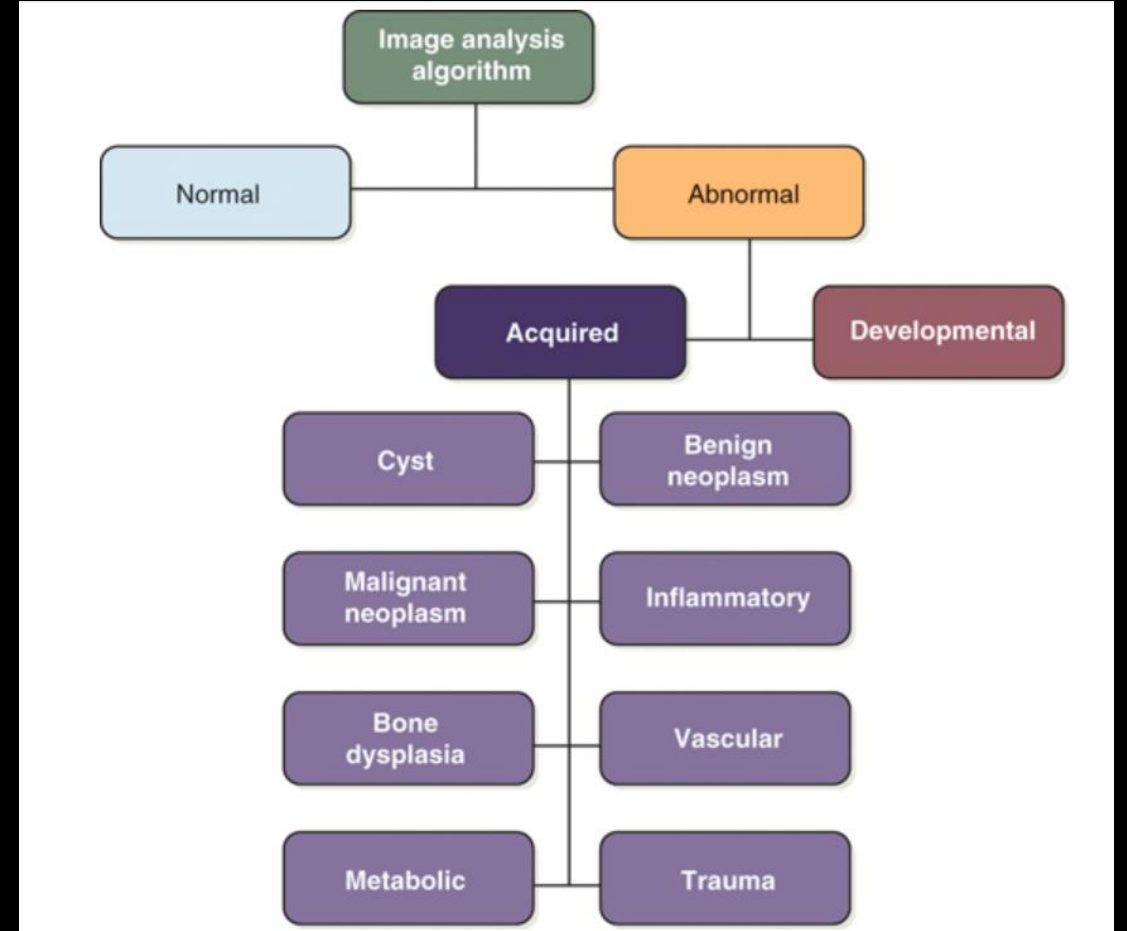
Step 4 :Analyse the effect on surrounding structure

- Teeth, lamina dura, PDL space
- IAC and mental foramen
- Maxillary antrum
- Surrounding bone density and trabecular pattern
- Outer cortical bone and periosteal reactions

Step 5 :Formulate a radiographic interpretation

Formulate a radiographic interpretation

- Normal or abnormal?
- Developmental or acquired?
- Classification
- Ways to proceed?

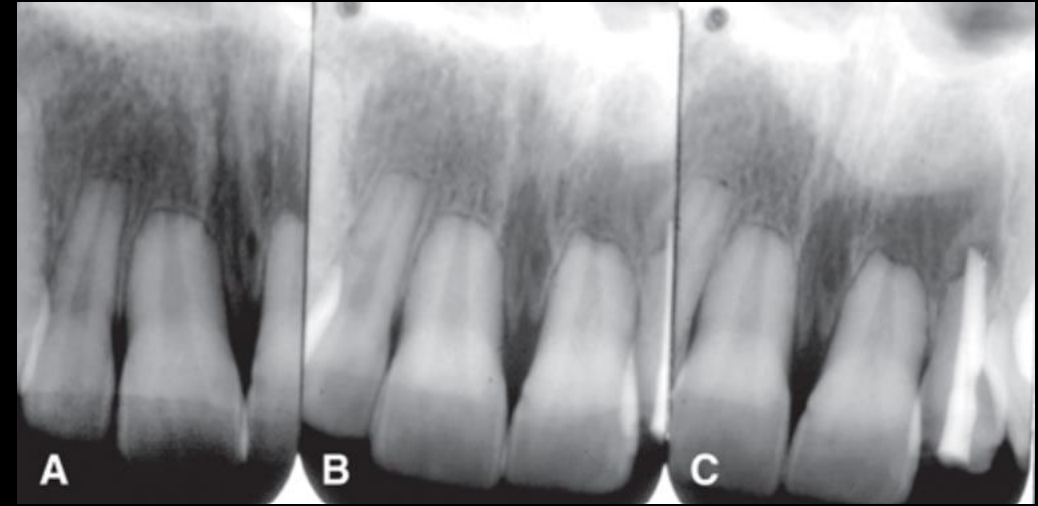


Decision 1 – Normal or Abnormal?

- **Key Question:** Is the finding a variation of normal or an abnormality?
- **Importance:**
 - Variations of normal do not require treatment.
 - Avoid unnecessary investigations.
- **Clinician's Role:**
 - Develop deep knowledge of normal anatomical variations.

Decision 2 – Developmental or Acquired?

- **Key Question:** Did the abnormality occur during development or was it acquired later?
- **Example:**
 - Short root:
 - Developmental → root canal not visible to apex.
 - Acquired (e.g., external resorption) → canal visible to apex.
- **Radiographic Clue:**
 - External resorption shows wide pulp chambers at apex.



- **Case – External Root Resorption**
- **Images A–C:**
 - PAs revealing external resorption of the maxillary incisors, which is an acquired abnormality because of the presence of the wide pulp chambers at the apex of the roots of the teeth.
- **Interpretation:**
 - Acquired abnormality due to external resorption.

Decision 3 – Disease Classification

- **Goal:** Identify the most likely disease category.
- **Categories Include:**
 - Inflammation
 - Cysts
 - Benign/Malignant Neoplasia
 - Bone Dysplasia
 - Vascular/Metabolic Disease
 - Trauma
- **Tip:** Use radiographic features + clinical info.

Integrating Clinical Information

- **Why Wait to Use Clinical Info?**
 - Prevents bias in image interpretation.
- **When to Use:**
 - After narrowing down radiographic possibilities.
- **Benefit:**
 - Helps refine to a differential diagnosis.

Decision 4 – How to Proceed

- **Next Steps May Include:**

- Further imaging
- Biopsy
- Observation
- Treatment

- **Example:**

- Suspected malignancy → advanced imaging + biopsy.
- Benign lesion → watchful waiting.



Diagnostic Imaging Report

- Structure and Guidelines for Radiographic Documentation and Communication



General Patient Information

- Patient name, age, sex
- Patient ID or registration number
- Referring clinician (if applicable)
- Date of report



Imaging Procedure

- Summary of imaging procedures performed
- Include date of examination
- For CBCT: field-of-view, slice thickness, spacing, image quality
- Example: Panoramic and occlusal images with contrast-enhanced CT on Feb 20, 2024

Clinical Information

- Optional section
- Brief summary of clinical findings
- Provided by referring clinician or examiner
- Example: Mass in floor of mouth, possible ranula; history of lymphoma

Findings

- Objective observations from images
- Describe:
 - Anatomic location
 - Lesion extent
 - Periphery and shape
 - Internal structure
 - Effects on surrounding structures
- No interpretation included

Interpretation

- Elucidate meaning of findings
- Combine with clinical history if available
- Provide diagnosis if possible
- If not, list differential diagnoses (limited to 1–2 categories)



Recommendations

- Suggest further studies if needed
- Include treatment advice if applicable
- Example: Recommend biopsy or follow-up imaging



Report Finalization

- Include name and signature of reporting clinician
- Ensure clarity and completeness
- Maintain professional and standardized format

RADIOGRAPHIC INTERPRETATION



Date Radiographs Taken: _____ Date of Initial Recordings: _____ Type of Radiographic Survey: _____
Student #: _____ Student Signature: _____ Faculty Signature: _____
Tooth Number: (O) Unerupted; (X) Missing; (#) Malposed
Crown: Root Ratio: (=) 1:1; (<) less than 1:1; (>) greater than 1:1
Calculus: (M) Mesial; (D) Distal; (s) Slight; (m) Moderate; (g) Gross
Alveolar Bone Loss: (H) Horizontal; (V) Vertical; (F) Furcation; (M) Mesial; (D) Distal
Periapical Abnormalities: (W) Widened PDL Space; (RO) Rarefying Osteitis; (SO) Sclerosing Osteitis
Apical Seal: Distance from apical seal to radiographic apex: (A) Within 1.5 – 2.0 mm; (Ux) underextended; (Ox) overextended
Obturation: (NV) No voids, dense obturation; (V) Voids – not acceptable, many voids and not uniform
Pulpal Abnormalities: (I) Increased Size; (R) Reduced Size; (St) Stone; (CC) Calcified Canals
Caries: (M) Mesial, (D) Distal, (C) Cervical, (O) Occlusal, (1, 2, 3, 4, 5) Radiographic Scoring System
Defective Restorations: (+) Overhang; (-) Sub Margin; (OM) Open Margin; (PC) Poor Contours
Occlusal Findings: (E) Even incisal edges/cusp tips; (Ue) Uneven incisal edges/cusp tips

Tooth #	Crown: Root Ratio	Calculus	Alveolar Bone Loss	Periapical Abnormalities	Pulp	Apical seal	Obturation	Caries	Defective restorations	Occlusion	Comments
1											
2											
3											
4A											
5B											
6C											
7D											
8E											
9F											
10G											
11H											
12I											
13J											
14											
15											
16											
17											
18											
19											
20K											
21L											
22M											
23N											
24O											
25P											
26Q											
27R											
28S											
29T											
30											
31											
32											

Body of Maxilla & Mandible (Comments):

